

1.4 - Wake Potential

Then, having the wake functions $\vec{w}$, one elegant (and powerful) further step is showing that these functions can be obtained from the derivatives of a scalar potential function W , which we'll call wake potential

REMINDER

In electromag.: $\vec{E} \rightarrow$ irrotational field $\Rightarrow \oint \vec{E} \cdot d\vec{l} = 0 \Rightarrow \int_A^B \vec{E} \cdot d\vec{l}$ is path independent

$\rightarrow$ enables defining $V(\vec{r}) \stackrel{\Delta}{=} - \int_O^{\vec{r}} \vec{E} \cdot d\vec{l} \Rightarrow - \int_A^B \vec{E} \cdot d\vec{l} = V_B - V_A$

$\Rightarrow \vec{E}(\vec{r}) = - \vec{\nabla} V(\vec{r})$ or simply $\vec{E} = - \vec{\nabla} V$ $\rightarrow$ gradient of a scalar potential

$\hookrightarrow \sim E_u(\vec{r}) = - \frac{\partial V}{\partial u}(\vec{r})$ $\hookrightarrow$ vector field

And just noticing:

$$E_u = \frac{F_u}{q} = \frac{1}{q} \frac{dp_u}{dt}; \quad w_u \sim \frac{v}{qQ} (\pm) dp_u$$

We want to find a $W(\vec{p}_s, \vec{p}_w, z)$ such that $w_u \propto \frac{\partial W}{\partial u}$

In particular, we'll have: $-\frac{\partial W}{\partial s} \sim$ (the minus sign being due to $w_{||} = -C \cdot \Delta p_s$, $C > 0$)

$$w_{||} = - \frac{\partial W}{\partial (-z)} = \frac{\partial W}{\partial z}; \quad \vec{w}_{\perp} = \vec{\nabla}_{w,\perp} W = \left(\frac{\partial}{\partial x_w}, \frac{\partial}{\partial y_w} \right) \cdot W$$

To find such potential, we start by writing the Lagrangian of the problem and, from the Euler-Lagrange equation $\left(\frac{d}{dt} \left(\frac{\partial L}{\partial \vec{v}} \right) = \vec{\nabla} L \right)$, obtain an expression for $\Delta \vec{p}$ ("a" $\vec{w}$), which will, in this specific case, be proportional to the gradient of a scalar function!

$\rightarrow$ the "a" are due to the minus sign in $w_{||} = -\Delta p_s$, so $\vec{w} \neq a \cdot \Delta \vec{p}$, meaning $\vec{w}$ and $\Delta \vec{p}$ are not literally proportional

Reminder on Lagrangian Mechanics

definition in classical mechanics

We begin with the Lagrangian: $L = T - V$

From which we can define the action:

$$S = \int L dt$$

Then, from the Principle of Least Action, we

know that any system will evolve in a way which extremizes $S \sim \delta S = 0$!

But what does $\delta S = 0$ mean? Say we begin at the optimal path $(q, \dot{q})$, so that S_i is the "extreme-valued" action. Then, when making a small deviation $(\delta q, \delta \dot{q})$ from that path, we should find $S_f = S_i + \delta S = S_i$,

i.e., $\delta S = 0$:

$$\begin{aligned} S(q + \delta q, \dot{q} + \delta \dot{q}) &= \int_{t_0}^{t_1} L(q + \delta q, \dot{q} + \delta \dot{q}, t) dt = \int_{t_0}^{t_1} \left[L(q, \dot{q}, t) + \frac{\partial L}{\partial q} \delta q + \frac{\partial L}{\partial \dot{q}} \delta \dot{q} \right] dt = \\ &= S(q, \dot{q}, t) + \int_{t_0}^{t_1} \frac{\partial L}{\partial q} \delta q dt + \int_{t_0}^{t_1} \frac{\partial L}{\partial \dot{q}} \frac{d}{dt} \delta q dt \\ &= S(q, \dot{q}, t) + \int_{t_0}^{t_1} \left[\frac{\partial L}{\partial q} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \right] \delta q dt \end{aligned}$$

1st order δq 's are small

$\delta \dot{q} = \delta \frac{dq}{dt} = \frac{d}{dt} \delta q$

$\delta q(t_0) = \delta q(t_1) = 0$

$\delta \dot{q} = \dot{q}_f - \dot{q}_i = \frac{d}{dt} (q_f - q_i) = \frac{d}{dt} \delta q$

$\Rightarrow \delta S = S(q + \delta q, \dot{q} + \delta \dot{q}) - S(q, \dot{q}, t) = \int_{t_0}^{t_1} \left[\frac{\partial L}{\partial q} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \right] \delta q dt$

If instead of $L(q, \dot{q}, t)$ we had $L(q_i, \dot{q}_i, t)$, we would have $\delta L = \sum_i \frac{\partial L}{\partial q_i} \delta q_i + \sum_i \frac{\partial L}{\partial \dot{q}_i} \delta \dot{q}_i$, and thus

$$\delta S = \int_{t_0}^{t_1} \delta L(q_i, \dot{q}_i, t) dt = \sum_i \left[\int_{t_0}^{t_1} \frac{\partial L}{\partial q_i} \delta q_i dt + \int_{t_0}^{t_1} \frac{\partial L}{\partial \dot{q}_i} \delta \dot{q}_i dt \right] = \int_{t_0}^{t_1} \sum_i \left[\left(\frac{\partial L}{\partial q_i} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) \right) \delta q_i \right] dt = 0$$

Euler-Lagrange equation *

Now, to have $\delta S = 0$ for any small δq_i , the

remaining of the integrands must all be zero everywhere, and thus

↳ independently

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0$$

V:

• Thus, by building the Lagrangian $L(q, \dot{q}, t)$ of a physical system, we can determine how $(q, \dot{q})$'s will evolve - i.e., the equations of movement of the system - by using $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) = \frac{\partial L}{\partial q_i}$.

• Illustrating the case for classical mechanics (1 particle):

$$\triangleright T = \frac{1}{2} m |\vec{\dot{q}}|^2 \Rightarrow L(\vec{q}, \vec{\dot{q}}, t) = \frac{1}{2} m |\vec{\dot{q}}|^2 - V(\vec{q})$$

$\vec{\dot{q}}^2 = \dot{q}_x^2 + \dot{q}_y^2 + \dot{q}_z^2$

Each coord. u :

$$\triangleright \text{Euler-Lagrange: } \frac{\partial L}{\partial \dot{q}_u} = m \dot{q}_u \Rightarrow \frac{d}{dt} (m \dot{q}_u) = \frac{\partial}{\partial q_u} (-V(\vec{q})) \Rightarrow m \ddot{q}_u = - \frac{\partial}{\partial q_u} V(\vec{q}) = F_u \quad ! \quad \sim \text{Newton's and Law}$$

Or in vec. terms:

$$\frac{d}{dt} \frac{\partial L}{\partial \vec{\dot{q}}} = \frac{\partial L}{\partial \vec{q}} = \frac{d}{dt} m \vec{\dot{q}} - \vec{\nabla} V \Rightarrow m \ddot{\vec{q}} = - \vec{\nabla} V \quad \sim \quad (m \ddot{q}_x, m \ddot{q}_y, m \ddot{q}_z) = - \left(\frac{\partial V}{\partial q_x}, \frac{\partial V}{\partial q_y}, \frac{\partial V}{\partial q_z} \right) =$$

→ Thus recovering the movement equation from Newtonian Mechanics!

$$= - \left(\frac{\partial}{\partial q_x}, \frac{\partial}{\partial q_y}, \frac{\partial}{\partial q_z} \right) V = - \vec{\nabla} V$$

• Expressing the E.L. equations for 3 coordinates (and using $\vec{\dot{q}} = \vec{v}$):

$$\boxed{\frac{d}{dt} \left(\frac{\partial L}{\partial \vec{v}} \right) = \vec{\nabla} L} \quad \rightarrow \quad \text{If } L = T - V: \quad \frac{d}{dt} (m \vec{v}) = \vec{\nabla} (-V) \Rightarrow \frac{d\vec{p}}{dt} = - \vec{\nabla} V \Rightarrow \frac{d\vec{p}}{dt} = \vec{F}$$

Newton's and Law very directly!

Now that this is clear, we can go back to the wake-fields context and find the Lagrangian of the witness particle (in whose $\Delta \vec{p} = "a" \vec{w}$ - we're interested in).

It is the Lagrangian of a relativistic charged particle moving through E.M. fields $\vec{E}$ and $\vec{B}$, i.e., under the influence of E.M. potentials ϕ and $\vec{A}$.

Let's build it in two parts:

- Relativistic free-particle term
- Electromagnetic interaction term



APPENDIX

(i) Lagrangian of a Relativistic Free-Particle

• In non-relativistic mechanics, we would have simply: $L = T = \frac{1}{2}mv^2$

• But in the relativistic context, we also have the requirement of $\int L dt$ being Lorentz invariant! ~ which is not the case for dt , ds or v ($\neq$ observers can disagree about them).

→ Thus, we should have something with the units of action (energy · time) which is also Lorentz invariant.

↳ The natural invariant along a particle's path is its proper time $d\tau$
 { actual time experienced by the particle
 independent of the observer's ref. frame
 fixed by the geom. of spacetime }

$$d\tau = \frac{dt}{\gamma}$$

 { time interval measured by a stationary observer's reference frame }
 { time registered by a clock which is at rest relative to the particle moving with }

→ Then, to adjust for the units,

the simplest L-inv. term with energy units

would be $A \cdot mc^2 \sim$ prop. to rest mass:

$$L = A mc^2 \sqrt{1 - v^2/c^2}$$

$$\Rightarrow \int L dt = \int A mc^2 \sqrt{1 - v^2/c^2} dt$$

Finally, to determ. A we can use Lagrangian formalism,

from which the canonical momentum is:

$$\vec{p}_{can} = \frac{\partial L}{\partial \vec{v}} = \frac{\partial}{\partial \vec{v}} \left(A mc^2 \sqrt{1 - v^2/c^2} \right) = A mc^2 \cdot \frac{-2\vec{v}/c^2}{2\sqrt{1 - v^2/c^2}} = -A \gamma m \vec{v}$$

Thus, for it to match the relativistic momentum, $\vec{p}_{rel} = \gamma m \vec{v}$, we need $A = -1$

And therefore, for a rel. free-particle the Lagrangian is:

$$L_{rel} = - \frac{mc^2}{\gamma} = - mc^2 \sqrt{1 - v^2/c^2}$$

and thus:
$$L_{rel} \stackrel{v \ll c}{\approx} - mc^2 \left(1 - \frac{v^2}{2c^2} \right) = - mc^2 + \frac{1}{2}mv^2$$

~ Notice that if $v \ll c \Rightarrow \frac{v^2}{c^2} \ll 1$:

$$(1 - u)^{1/2} = 1 + \frac{1}{2}(1 - u)^{-1/2} \cdot u + \dots \stackrel{u \ll 1}{\approx} 1 - \frac{u}{2}$$

at $u=0$
 $\Rightarrow \frac{\partial L}{\partial \vec{v}} = m \vec{v} = \vec{p}_{classical}$
 i.e. doesn't affect the eq's of motion:

(ii) Electromagnetic Interaction Lagrangian Term

• Forgetting relativity for a moment, ^(v < c) we want to write a Lagrangian for which the Euler-Lagrange eq. gives a force term which is precisely the Lorentz Force, $\vec{F}_L = q(\vec{E} + \vec{v} \times \vec{B})$.

• We remember that $\vec{E}$ and $\vec{B}$ (due to satisfying $\vec{\nabla} \times \vec{E} = 0$ and $\vec{\nabla} \cdot \vec{B} = 0$) can be written in terms of the electric and magnetic potentials: $\vec{E} = -\vec{\nabla}\phi$, $\vec{B} = \vec{\nabla} \times \vec{A}$.
 $\Rightarrow \vec{E} = -\vec{\nabla}\phi \sim$ "electrostatic case"
 $\phi = \phi(\vec{r}) \sim \text{scal. 3D func}$
 $\vec{A} = \vec{A}(\vec{r}) \sim \text{vec. 3D field}$

• To build an intuition, we start with the case where only an electric force is acting on the particle (purely electrostatic $\sim \vec{B} = 0 \Rightarrow \vec{E} = -\vec{\nabla}\phi$)

$\rightarrow$ The potential energy is then: $V_{el} = q \cdot \phi(\vec{r})$
 Euler-Lagrange: $\frac{d}{dt} \left(\frac{\partial L}{\partial \vec{v}} \right) = \vec{\nabla} V$

$\rightarrow$ Plugging into the Lagrangian: $L = \frac{1}{2} m |\vec{v}|^2 - q\phi$
 $\Rightarrow \frac{d}{dt} \vec{p} = -\vec{\nabla} q\phi \Rightarrow \vec{F}_E = -\vec{\nabla} V_{el}$ ✓

From now on we should make the Maxwell correction $\vec{E} = -\vec{\nabla}\phi - \frac{\partial \vec{A}}{\partial t}$

• Now, to account for the $q\vec{v} \times \vec{B}$ of the Lorentz force, we see that

$\vec{v} \times \vec{B} = \vec{v} \times \vec{\nabla} \times \vec{A} = \underbrace{(\vec{\nabla}(\vec{v} \cdot \vec{A}) - \vec{A}(\vec{\nabla} \cdot \vec{v}))}_{\text{B(A.C) - C(A.B) rule w/ operator abstractions for and (but)}}$, and based on the 1st term, make the guess:

$\rightarrow V = q\phi - q\vec{v} \cdot \vec{A}$

$\rightarrow$ E.L.: $\frac{d}{dt} \left(\frac{\partial L}{\partial \vec{v}} \right) - \vec{\nabla} L = \frac{d}{dt} (m\vec{v} + q\vec{A}) - q\vec{\nabla}(-\phi + \vec{v} \cdot \vec{A}) =$

$\rightarrow$ Lagrangian: $L = \frac{1}{2} m |\vec{v}|^2 - q\phi + q\vec{v} \cdot \vec{A}(\vec{r}) = \frac{d\vec{p}}{dt} + q \frac{d\vec{A}}{dt} + q\vec{v}\phi - q\vec{v}(\vec{v} \cdot \vec{A}) = 0$ *

But notice that $\frac{d}{dt} \vec{A}(\vec{r}) = \frac{\partial \vec{A}}{\partial t} + \frac{\vec{r}}{\partial t} \cdot \frac{\partial \vec{A}}{\partial \vec{r}} = \frac{\partial \vec{A}}{\partial t} + (\vec{v} \cdot \vec{\nabla}) \vec{A}$
 $\sim \frac{dA_x(\vec{r}, t)}{dt} = \frac{\partial A_x}{\partial t} + \frac{\partial A_x}{\partial x} \frac{dx}{dt} + \frac{\partial A_x}{\partial y} \frac{dy}{dt} + \frac{\partial A_x}{\partial z} \frac{dz}{dt} = \frac{\partial A_x}{\partial t} + (\vec{v} \cdot \vec{\nabla}) A_x$

And thus:

* $\Rightarrow \frac{d\vec{p}}{dt} = +q \left[\underbrace{\frac{\partial \vec{A}}{\partial t} - (\vec{v} \cdot \vec{\nabla}) \vec{A}}_{= \vec{\nabla} \times (\vec{v} \times \vec{A})!} + \vec{\nabla}(\vec{v} \cdot \vec{A}) - \vec{\nabla}\phi \right] = q \left[\underbrace{\left(\vec{\nabla}\phi + \frac{\partial \vec{A}}{\partial t} \right)}_{\vec{E}} + \underbrace{\vec{v} \times (\vec{\nabla} \times \vec{A})}_{\vec{v} \times \vec{B}} \right]$

$\Rightarrow \frac{d\vec{p}}{dt} = q(\vec{E} + \vec{v} \times \vec{B}) = \vec{F}_L$

$\sim$ properly giving Newton's 2nd Law for a non-relativistic charged particle w/ velocity $\vec{v}$ under effect of the Lorentz force. ✓

Thus, the full Lagrangian for the witness particle / relativistic charged particle in electromagnetic potentials ϕ and A is:

$$L = -mc^2 \sqrt{1 - v^2/c^2} + q \vec{A} \cdot \vec{v} - q\phi, \quad (I)$$

such that:

$$S = \int L dt = \underbrace{-mc^2 \int \frac{dt}{\gamma}}_{S_{rel}} + \underbrace{q \int (\vec{A} \cdot \vec{v} - \phi) dt}_{S_{int.}}$$

$\bullet dS_{rel} = -mc^2 d\tau$
 $\sim$ a particle with mass accumulates action proportional to its proper time $d\tau$
 $\bullet dS_{int.} = q(\vec{A} \cdot d\vec{r} - \phi dt)$
 $\sim$ a charged particle accumulates action by moving through electromagnetic potentials

where $\vec{A}$ and ϕ are the electromag. potentials defined by the ME for the source particle together with the definitions $\vec{E} = -\vec{\nabla}\phi - \frac{\partial \vec{A}}{\partial t}$ and $\vec{B} = \vec{\nabla} \times \vec{A}$.
 which gives $\vec{E}$ and $\vec{B}$

Then, applying the Euler-Lagrange equations to (I):

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \vec{v}} \right) - \vec{\nabla} L = \frac{d}{dt} \left(-mc^2 \frac{-\vec{v}/c^2}{\sqrt{1-v^2/c^2}} + q\vec{A} \right) - q\vec{\nabla} (\vec{A} \cdot \vec{v} - \phi) = \frac{d}{dt} (\gamma m \vec{v} + q\vec{A}) - q\vec{\nabla} (A_z v_z - \phi)$$

$v_x, v_y \ll v_z$

$$\Rightarrow \frac{d}{dt} (\vec{p}_{rel} + q\vec{A}) = q\vec{\nabla} (A_z v_z - \phi)$$

Integrating both sides:

$$\left[\vec{p} + q\vec{A} \right]_{t=-\infty}^{t=+\infty} = \underbrace{\Delta \vec{p}}_{\substack{\text{momentum} \\ \text{change}}} = q \cdot \int_{-\infty}^{+\infty} \vec{\nabla} (A_z v_z - \phi) dt = q \vec{\nabla} \int_{-\infty}^{+\infty} (A_z v_z - \phi) dt$$

assumption that x, y are fixed with t
 gradient of a scalar function
 as we derived!
 $\left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right)$

Remembering that $\vec{W} = \frac{c}{qQ} (\Delta p_x, \Delta p_y, -\Delta p_z)$ and seeing that we can write $\Delta \vec{p} = \frac{qQ}{c} \vec{\nabla}_R W$, where

$$\vec{R} = (x_w, y_w, -z) \Rightarrow \Delta \vec{p} = \frac{qQ}{c} \left(\frac{\partial W}{\partial x_w}, \frac{\partial W}{\partial y_w}, -\frac{\partial W}{\partial z} \right), \text{ we have:}$$

$$\Rightarrow (w_x, w_y, w_z) = \frac{c}{qQ} \cdot \frac{qQ}{c} \left(\frac{\partial W}{\partial x_w}, \frac{\partial W}{\partial y_w}, -\frac{\partial W}{\partial z} \right) \Rightarrow \begin{cases} w_x = \frac{\partial W}{\partial x_w} \\ w_y = \frac{\partial W}{\partial y_w} \\ w_z = -\frac{\partial W}{\partial z} \end{cases} !$$

So, in short:

$$\Delta \vec{p} = \vec{\nabla} \left(\underbrace{\frac{q}{c} \int_{-\infty}^{\infty} |A_s v_s - \phi| dt}_{f(\vec{r})} \right) = \frac{q \cdot Q}{c} (w_x, w_y, -w_z)$$

$\text{kg m/s} \quad \text{kg m}^2/\text{s}^2 \cdot 1/c^2 = \mathcal{T}/c^2 = V/c$

$$\Rightarrow (w_x, w_y, -w_z) = \frac{c}{qQ} \cdot \vec{\nabla} f = \vec{\nabla} \left(\frac{c}{qQ} \int_{-\infty}^{\infty} |A_s v_s - \phi| dt \right) = \vec{\nabla}_R W(\vec{\rho}_s, \vec{\rho}_w, z)$$

$\left(\frac{\partial}{\partial x_w}, \frac{\partial}{\partial y_w}, \frac{\partial}{\partial z} \right) = \left(\frac{\partial}{\partial x_w}, \frac{\partial}{\partial y_w}, \frac{\partial}{\partial(z-z)} \right)$
 $\sim z$

And thus:

$$(w_x, w_y, -w_z) = \vec{\nabla}_R W \Rightarrow \begin{cases} (w_x, w_y) = \vec{w}_\perp = \vec{\nabla}_{w,\perp} W \\ w_z = -\frac{\partial}{\partial z} W = \frac{\partial}{\partial z} W \end{cases}$$

Furthermore, this potential formalism enables directly obtaining the so-called Panofsky-Wenzel theorem from the second cross derivatives of W :

$$\frac{\partial \vec{w}_\perp}{\partial z} = \frac{\partial}{\partial z} (\vec{\nabla}_{w,\perp} W) = \vec{\nabla}_{w,\perp} \left(\frac{\partial}{\partial z} W \right) = \vec{\nabla}_{w,\perp} w_z \rightarrow \frac{\partial \vec{w}_\perp}{\partial z} = \vec{\nabla}_{w,\perp} w_z \quad (\text{II})$$

↳ This property of the wakes is very useful as it enables calculating $\vec{w}_\perp$ from w_z by

a simple integration of (II):

$$\vec{w}_\perp = \int_{-\infty}^z \vec{\nabla}_{w,\perp} w_z dz^* \quad \text{Furthermore, } w_z = -\frac{c}{qQ} \Delta p_s = -\frac{c}{qQ} \int_{-\infty}^{\infty} \underbrace{(F_L)}_{\text{impulse approx.}} dt = -\frac{c}{Q} \int_{-\infty}^{\infty} dt E_s|_{s=ct-z}$$

$\vec{F}_L = q[E_s \hat{s} + (\epsilon_x - vB_y)\hat{x} + (\epsilon_y + vB_x)\hat{y}]$

⇒ All wake functions can be calculated from E_s !

$$w_z(\vec{\rho}_s, \vec{\rho}_w, z) = -\frac{c}{Q} \int_{-\infty}^{\infty} dt E_s(\vec{\rho}_s, \vec{\rho}_w, z, t) \Big|_{s=ct-z}; \quad \vec{w}_\perp = \int_{-\infty}^z \vec{\nabla}_{w,\perp} w_z dz^* !$$

Another important property of the wake potential W is being an harmonic function in the transverse coordinates x and y [if $\beta = 1 \sim v = \beta c$]*:

$$\nabla_{\perp}^2 W = \frac{\partial^2 W}{\partial x^2} + \frac{\partial^2 W}{\partial y^2} = 0 \quad \sim \quad \frac{\partial W_x}{\partial x} = -\frac{\partial W_y}{\partial y}$$

To prove this, we first remember that, from the ME in free space, the potentials ϕ and $\vec{A}$ satisfy the wave equation:

$$\nabla^2 \vec{A} - \overset{1/c^2}{\epsilon_0} \frac{\partial^2 \vec{A}}{\partial t^2} = 0; \quad \nabla^2 \phi - \epsilon_0 \frac{\partial^2 \phi}{\partial t^2} = 0$$

Now, since $W = q \int_{-\infty}^{\infty} (A_s v - \phi) dt$, we have:

$$\nabla^2 W = \nabla_{\perp}^2 W + \frac{\partial^2 W}{\partial z^2} = \nabla^2 \left(q \int_{-\infty}^{\infty} (A_s v - \phi) dt \right) = q \int_{-\infty}^{\infty} (c \nabla^2 A_s - \nabla^2 \phi) dt = q \int_{-\infty}^{\infty} \frac{1}{c^2} \frac{\partial^2}{\partial t^2} (v A_s - \phi) dt$$

$$\Rightarrow \nabla_{\perp}^2 W = q \int_{-\infty}^{\infty} \left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial z^2} \right) (v A_s - \phi) dt$$

$$\text{Furthermore, } \left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial z^2} \right) = \left(\frac{1}{c} \frac{\partial}{\partial t} + \frac{\partial}{\partial z} \right) \cdot \left(\frac{1}{c} \frac{\partial}{\partial t} - \frac{\partial}{\partial z} \right), \text{ and } z = \beta c t - z \Rightarrow \frac{d}{dt} = \left(\frac{\partial}{\partial t} + \frac{\partial z}{\partial t} \frac{\partial}{\partial z} \right) = \beta c \left(\frac{1}{\beta c} \frac{\partial}{\partial t} + \frac{\partial}{\partial z} \right),$$

meaning that, for $\beta = 1$:

$$\nabla_{\perp}^2 W = q \int_{-\infty}^{\infty} c \frac{d}{dt} \left(\frac{1}{c} \frac{\partial}{\partial t} - \frac{\partial}{\partial z} \right) (c A_s - \phi) dt = q c \cdot \left[\left(\frac{1}{c} \frac{\partial}{\partial t} - \frac{\partial}{\partial z} \right) (c A_s - \phi) \right]_{t=-\infty}^{t=\infty} = 0$$

~ since $\vec{A}$ and ϕ vanish at $\pm \infty$

$$\therefore \nabla_{\perp}^2 W = \frac{\partial^2 W}{\partial x^2} + \frac{\partial^2 W}{\partial y^2} = 0$$